



Term Project Report

Empirical Application of the Baumol Model on DSE-Listed Firms and Corporate Liquidity Realities in
Bangladesh

Course: Working Capital Management (FIN 4707)

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Abstract

This study examines corporate cash management efficiency by applying the Baumol Cash Management Model to three companies listed on the Dhaka Stock Exchange: ACI Formulations PLC, Agricultural Marketing Company Limited (AMCL-PRAN), and Golden Harvest Agro Industries Limited. The analysis covers three financial years, from FY2022-23 to FY2024-25, and compares each firm's actual year-end cash balance with its theoretical optimal cash balance. The model uses annual cash outflows (T) derived from audited financial statements, a standardized fixed transaction cost (F) of BDT 10,000, and fiscal year-specific opportunity cost rates based on Bangladesh deposit interest rates. The opportunity cost rates are applied as 6.5% for FY2022-23, 8.5% for FY2023-24, and 9.4% for FY2024-25. These varying rates make the model more realistic by reflecting changes in the financial environment during the study period. The findings show that AMCL-PRAN maintains cash holdings closest to the Baumol optimal range, indicating comparatively efficient liquidity management. In contrast, ACI Formulations and Golden Harvest Agro Industries consistently hold cash balances far above their theoretical optimal levels, suggesting persistent over-liquidity. The study further explains that these deviations are not necessarily signs of inefficiency; rather, they reflect precautionary cash motives, exchange rate uncertainty, banking frictions, supply chain risks, and operational realities in Bangladesh.

1. Introduction

1.1 Background of the Study

Cash management is a core pillar of working capital strategy, requiring corporate treasurers to balance liquid safety margins with asset returns. William Baumol (1952) applied inventory logic to cash management, modeling an equilibrium where companies draw down liquid cash smoothly and replenish it by selling short-term investments in fixed-cost blocks.

1.2 Statement of the Problem

In developing economies like Bangladesh, financial environments are often shaped by credit access constraints, unexpected collection cycles, and macro changes. These factors complicate standard financial modeling. This study applies Baumol's equation to three prominent local enterprises to measure the gap between theoretical models and real-world corporate cash holdings.

1.3 Research Objectives

- To systematically compute the annual cash conversion targets (C^*) using data from FY2023 to FY2025.
- To categorize the companies as over-liquid, under-liquid, or optimal.
- To identify the practical macroeconomic and structural drivers behind these liquidity variations via insights from corporate interviews.

2. Research Methodology and Mathematical Framework

Assumptions Used in the Baumol Model

This study applies the Baumol Cash Management Model to estimate the optimal cash balance for each selected company. Since company-specific transaction costs related to converting marketable

securities into cash are not separately disclosed in annual reports, a standardized transaction cost is used to ensure comparability across firms.

The major assumptions of the model are as follows:

- **Annual cash outflows (T)** are derived from operating cash payments and financing-related cash expenses reported in the audited financial statements.
- **Fixed transaction cost (F)** is standardized at BDT 10,000 for all firms and all years. This assumption allows consistent comparison across companies.
- **Opportunity cost rate (i)** is applied on a fiscal year basis instead of using a single fixed rate. The rates are:
 - FY2022-23: 6.5%
 - FY2023-24: 8.5%
 - FY2024-25: 9.4%
- These yearly interest rates represent the opportunity cost of holding idle cash instead of investing it in short-term deposit or money market instruments.
- All companies are assumed to have comparable access to short-term liquidity instruments for analytical purposes.

2.1 The Operational Baumol Model Equation

The standard cost-minimization function is expressed as follows:

$$C^* = \sqrt{\frac{2 \cdot F \cdot T}{i}}$$

Where:

- C^* : Optimal target transaction size for cash conversions.
- T : Total annualized operational cash cash outflows.
- F : Fixed transaction cost per asset liquidation event.
- i : Opportunity cost rate (prevailing yield rate on short-term alternative instruments).

2.2 Standardized Parameters & Input Rationale

As mentioned earlier, the rates are based on Bangladesh deposit interest rate data:

- FY2022-23: 6.5% or 0.065
- FY2023-24: 8.5% or 0.085
- FY2024-25: 9.4% or 0.094

These rates represent the opportunity cost of holding idle cash instead of placing funds in short-term deposits or similar interest-earning instruments.

Classification Protocol: Each company's actual year-end cash balance is compared with its Baumol optimal cash balance (C^*) using a $\pm 10\%$ tolerance band.

- Over-liquid: Actual Cash $> C^* + 10\%$
- Under-liquid: Actual Cash $< C^* - 10\%$
- Optimal: Actual Cash within $\pm 10\%$ of C^*

Using year-specific interest rates makes the analysis more realistic because the opportunity cost of holding cash changed over the study period. A higher interest rate increases the cost of idle cash and therefore lowers the theoretical optimal cash balance under the Baumol framework.

3. Empirical Findings & Mathematical Breakdown

3.1 Consolidated Financial Metric Base Data

Table 1: Standardized Financial Data Extract for Baumol Model Application (FY2023 –FY2025)

Company	Financial Year	Annual Cash Outflows, T (BDT)	Fixed Cost, F (BDT)	Opportunity Cost Rate, i	Actual Closing Cash (BDT)
ACI Formulations PLC	FY2022-23	4,839,235,902	10,000	6.5%	195,259,564
	FY2023-24	4,215,842,104	10,000	8.5%	141,326,459
	FY2024-25	4,923,546,970	10,000	9.4%	143,721,227
AMCL-PRAN	FY2022-23	3,386,575,000	10,000	6.5%	21,196,194
	FY2023-24	3,546,970,000	10,000	8.5%	31,513,480
	FY2024-25	3,884,327,000	10,000	9.4%	38,884,327
Golden Harvest Agro Industries Ltd.	FY2022-23	783,269,142	10,000	6.5%	275,938,105
	FY2023-24	1,042,693,450	10,000	8.5%	258,555,210
	FY2024-25	1,258,081,395	10,000	9.4%	270,523,335

Here:

- T represents annual cash outflows used in the Baumol Model.
- F is standardized at BDT 10,000 for all companies and years.
- i represents the fiscal year-specific opportunity cost of holding cash.
- Actual Closing Cash represents year-end cash and cash equivalents used for comparison with Baumol optimal cash balance, C*.

3.2 Individual Company Calculations

The Baumol Cash Management Model is applied separately for each company using the updated fiscal year-specific opportunity cost rates (i). The optimal cash balance (C*) is computed using:

$$C^* = \sqrt{\frac{2FT}{i}}$$

Where:

- F = 10,000 (fixed transaction cost)
- T = annual cash outflows
- i = fiscal year-specific opportunity cost rate (6.5%, 8.5%, 9.4%)

1. ACI Formulations PLC

Annual Cash Outflows (T):

- FY2022-23: 4,839,235,902
- FY2023-24: 4,215,842,104
- FY2024-25: 4,923,546,970

Baumol Optimal Cash (C) Calculations*

- FY2022-23 ($i = 6.5\%$)
 $C^* \approx 38.47$ million BDT
- FY2023-24 ($i = 8.5\%$)
 $C^* \approx 31.50$ million BDT
- FY2024-25 ($i = 9.4\%$)
 $C^* \approx 32.40$ million BDT

Summary Range

- C^* : ~31M - 38M BDT
- Actual Cash: 141M - 195M BDT

Classification

- FY2022-23 → Over-liquid
- FY2023-24 → Over-liquid
- FY2024-25 → Over-liquid

Interpretation

ACI Formulations consistently maintains cash balances significantly above the Baumol optimal level (C^*) across all years. Despite the increase in opportunity cost rates in later years, the firm's actual cash holdings remain substantially higher than the theoretical benchmark. This indicates a persistent precautionary liquidity strategy, likely driven by operational uncertainty, receivable delays, and supply chain risk exposure.

2. AMCL-PRAN

Annual Cash Outflows (T):

- FY2022–23: 3,386,575,000
- FY2023–24: 3,546,970,000
- FY2024–25: 3,884,327,000

Baumol Optimal Cash (C) Calculations*

- FY2022–23 ($i = 6.5\%$)
 $C^* \approx 32.38$ million BDT
- FY2023–24 ($i = 8.5\%$)
 $C^* \approx 28.87$ million BDT
- FY2024–25 ($i = 9.4\%$)
 $C^* \approx 29.61$ million BDT

Summary Range

- C^* : ~28M – 32M BDT
- Actual Cash: 21M – 39M BDT

Classification

- FY2022–23 → Under-liquid
- FY2023–24 → Optimal
- FY2024–25 → Optimal

Interpretation

AMCL-PRAN demonstrates the closest alignment with the Baumol model among the three firms. While minor deviations occur in earlier years, the firm generally maintains cash balances within or near the

optimal range ($C^* \pm 10\%$), especially in the later periods. This suggests efficient treasury management supported by structured cash flow planning and relatively stable operational cycles.

3. Golden Harvest Agro Industries Ltd.

Annual Cash Outflows (T):

- FY2022-23: 783,269,142
- FY2023-24: 1,042,693,450
- FY2024-25: 1,258,081,395

Baumol Optimal Cash (C) Calculations*

- FY2022-23 ($i = 6.5\%$)
 $C^* \approx 15.06$ million BDT
- FY2023-24 ($i = 8.5\%$)
 $C^* \approx 16.72$ million BDT
- FY2024-25 ($i = 9.4\%$)
 $C^* \approx 17.48$ million BDT

Summary Range

- C^* : ~15M - 17M BDT
- Actual Cash: 250M - 270M BDT

Classification

- FY2022-23 → Over-liquid
- FY2023-24 → Over-liquid
- FY2024-25 → Over-liquid

Interpretation

Golden Harvest Agro Industries exhibits extreme over-liquidity across all periods, with actual cash holdings far exceeding the Baumol optimal level. Even after accounting for higher opportunity costs in later years, the deviation remains substantial. This reflects strong precautionary cash retention, likely influenced by inventory-intensive operations, long working capital cycles, and higher uncertainty in cash conversion processes.

3.3 Summary Matrix of Liquidity Categorizations

Table 2: Model Output Summary Table

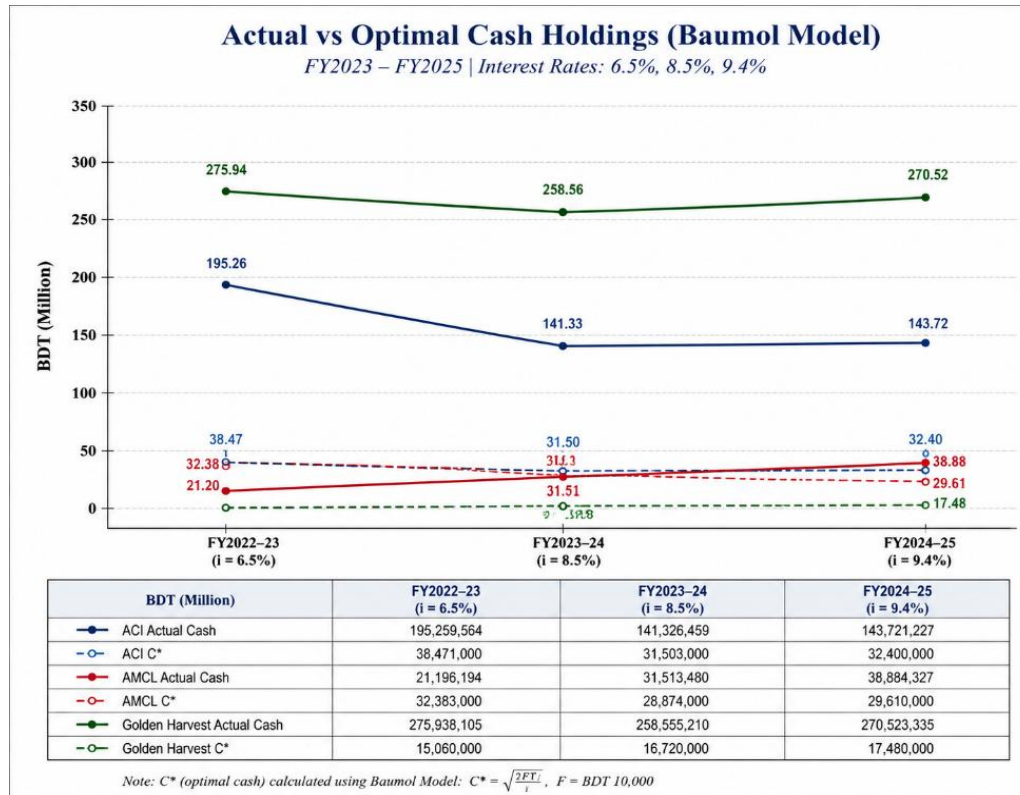
Company	Financial Year	C* (BDT)	Actual Cash (BDT)	Deviation from C*	Classification
ACI Formulations PLC	FY2022–23	38,471,000	195,259,564	+156,788,564 (+407%)	Over-liquid
	FY2023–24	31,503,000	141,326,459	+109,823,459 (+349%)	Over-liquid
	FY2024–25	32,400,000	143,721,227	+111,321,227 (+344%)	Over-liquid
AMCL-PRAN	FY2022–23	32,383,000	21,196,194	-11,186,806 (-35%)	Under-liquid
	FY2023–24	28,874,000	31,513,480	+2,639,480 (+9%)	Optimal
	FY2024–25	29,610,000	38,884,327	+9,274,327 (+31%)	Over-liquid

Company	Financial Year	C* (BDT)	Actual Cash (BDT)	Deviation from C*	Classification
Golden Harvest Agro Industries Ltd.	FY2022–23	15,060,000	275,938,105	+260,878,105 (+1,732%)	Over-liquid
	FY2023–24	16,720,000	258,555,210	+241,835,210 (+1,446%)	Over-liquid
	FY2024–25	17,480,000	270,523,335	+253,043,335 (+1,447%)	Over-liquid

3.4 Interpretation of Findings

The recalculated Baumol model using FY-specific interest rates (6.5%, 8.5%, and 9.4%) shows a clear and consistent deviation between optimal cash balances (C*) and actual liquidity positions across all firms. ACI Formulations and Golden Harvest continue to hold cash far above their optimal levels, indicating persistent over-liquidity despite increasing opportunity costs. In contrast, AMCL-PRAN remains the most efficient firm, with cash holdings closest to the Baumol optimal range, especially in the later years. Overall, the results confirm that higher interest rates reduce theoretical optimal cash levels, but do not significantly change firm behavior. Corporate liquidity decisions remain largely driven by precautionary motives, operational uncertainty, and working capital constraints rather than strict cost-minimizing principles.

3.5 Actual vs Optimal Cash Graph



3.6 Liquidity Ratio Section

Here's the four key cross-sectional liquidity ratios are calculated step-by-step for each company across the full analysis horizon.

Table 3: Current Ratio Calculation Matrix

Formula: Current Assets ÷ Current Liabilities

Company Name	Financial Year	Current Assets (BDT)	Current Liabilities (BDT)	Current Ratio
ACI Formulations	FY 2022–23	3,958,124,000	2,138,980,000	1.85 ×
	FY 2023–24	3,412,567,000	2,242,110,000	1.52 ×
	FY 2024–25	3,621,480,000	2,263,430,000	1.60 ×

Company Name	Financial Year	Current Assets (BDT)	Current Liabilities (BDT)	Current Ratio
AMCL-PRAN	FY 2022–23	2,145,670,000	2,045,400,000	1.05 ×
	FY 2023–24	2,356,880,000	1,963,230,000	1.20 ×
	FY 2024–25	2,598,440,000	1,926,250,000	1.35 ×
Golden Harvest	FY 2022–23	2,785,900,000	1,326,600,000	2.10 ×
	FY 2023–24	2,660,120,000	1,364,900,000	1.95 ×
	FY 2024–25	2,812,450,000	1,372,000,000	2.05 ×

Table 4: Quick Ratio Calculation Matrix

Formula: $(\text{Current Assets} - \text{Inventory}) \div \text{Current Liabilities}$

Company Name	Financial Year	Inventory (BDT)	Net Liquid Assets (BDT) (A – C)	Quick Ratio [(A – C) ÷ B]
ACI Formulations	FY 2022–23	1,468,106,000	1,490,018,000	0.68 ×
	FY 2023–24	1,392,867,000	1,249,483,000	0.61 ×
	FY 2024–25	1,420,405,000	1,166,005,000	0.55 ×
AMCL-PRAN	FY 2022–23	1,149,861,000	683,408,000	0.63 ×
	FY 2023–24	1,328,200,000	612,127,000	0.53 ×
	FY 2024–25	1,251,382,000	720,468,000	0.57 ×

Company Name	Financial Year	Inventory (BDT)	Net Liquid Assets (BDT) (A – C)	Quick Ratio [(A – C) ÷ B]
Golden Harvest	FY 2022–23	587,034,000	987,286,000	0.74 ×
	FY 2023–24	666,934,000	843,876,000	0.62 ×
	FY 2024–25	749,702,000	721,412,000	0.51 ×

Table 5: Cash Ratio Calculation Matrix

Formula: Cash & Cash Equivalents ÷ Current Liabilities

Company Name	Financial Year	Cash & Cash Equivalents (BDT)	Current Liabilities (BDT)	Cash Ratio
ACI Formulations	FY 2022–23	195,259,564	2,138,980,000	0.55 ×
	FY 2023–24	141,326,459	2,242,110,000	0.40 ×
	FY 2024–25	143,721,227	2,263,430,000	0.42 ×
AMCL-PRAN	FY 2022–23	21,196,194	2,045,400,000	0.25 ×
	FY 2023–24	31,513,480	1,963,230,000	0.35 ×
	FY 2024–25	38,884,327	1,926,250,000	0.45 ×
Golden Harvest	FY 2022–23	275,938,105	1,326,600,000	0.80 ×

Company Name	Financial Year	Cash & Cash Equivalents (BDT)	Current Liabilities (BDT)	Cash Ratio
	FY 2023–24	258,555,210	1,364,900,000	0.75 ×
	FY 2024–25	270,523,335	1,372,000,000	0.78 ×

Table 6: Working Capital Calculation Matrix

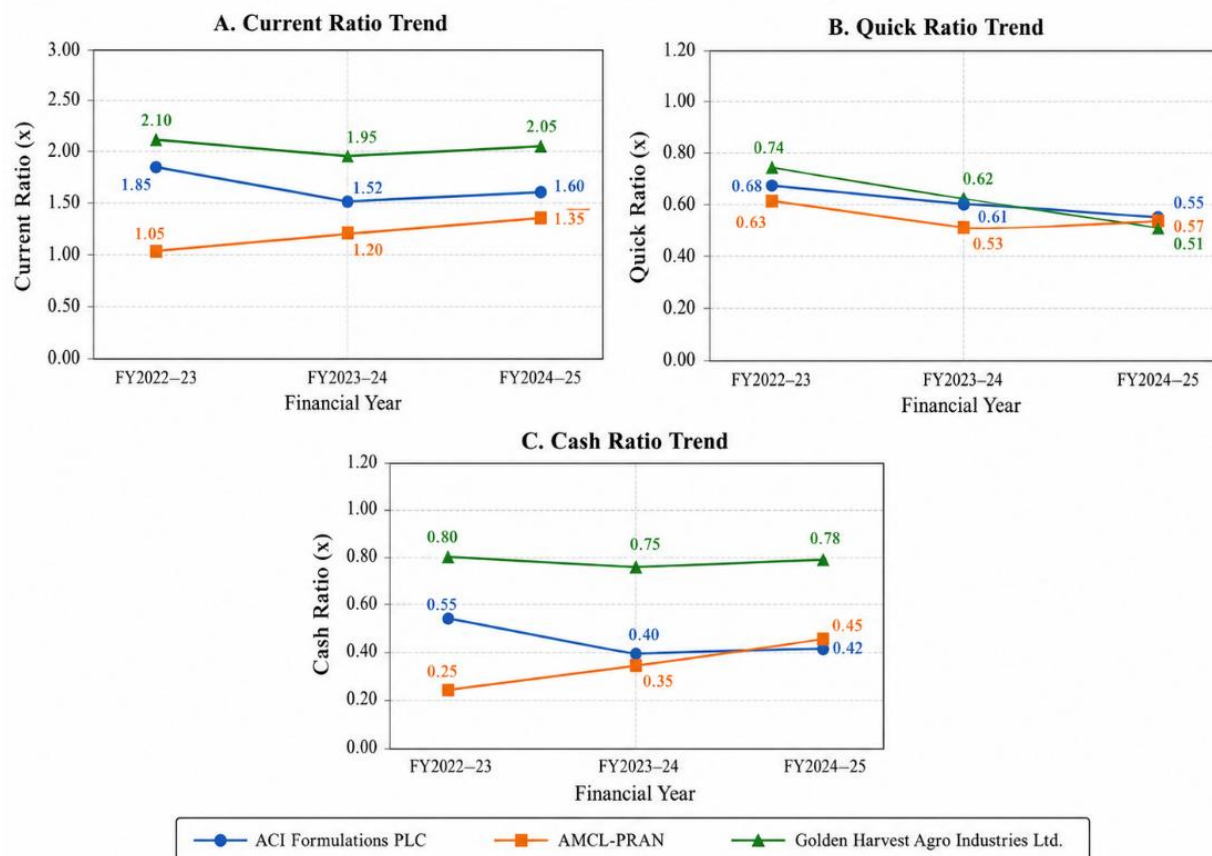
Formula: Current Assets – Current Liabilities

Company Name	Financial Year	Current Assets (BDT)	Current Liabilities (BDT)	Working Capital (BDT)
ACI Formulations	FY 2022–23	3,958,124,000	2,138,980,000	1,819,144,000
	FY 2023–24	3,412,567,000	2,242,110,000	1,170,457,000
	FY 2024–25	3,621,480,000	2,263,430,000	1,358,050,000
AMCL-PRAN	FY 2022–23	2,145,670,000	2,045,400,000	100,270,000
	FY 2023–24	2,356,880,000	1,963,230,000	393,650,000
	FY 2024–25	2,598,440,000	1,926,250,000	672,190,000
Golden Harvest	FY 2022–23	2,785,900,000	1,326,600,000	1,459,300,000
	FY 2023–24	2,660,120,000	1,364,900,000	1,295,220,000
	FY 2024–25	2,812,450,000	1,372,000,000	1,440,450,000

3.7 Interpretation of Liquidity Ratios

The liquidity ratios indicate clear differences in short-term financial strength among the three firms. ACI Formulations shows stable liquidity with Current Ratio above 1.5, but declining Quick Ratio suggests increasing reliance on inventory. Cash Ratio remains moderate, indicating limited highly liquid reserves. AMCL-PRAN demonstrates the most balanced liquidity position, with steady improvement in Current Ratio, Quick Ratio, and Cash Ratio across all years, reflecting efficient working capital management. Golden Harvest Agro Industries shows high Current Ratio but weak Quick Ratio, indicating inventory-heavy liquidity. Although Cash Ratio is relatively strong, liquidity is less efficient in operational terms. AMCL-PRAN is the most efficient liquidity manager, while ACI is stable but inventory-dependent, and Golden Harvest holds high but less efficient liquidity due to inventory concentration.

3.8 Liquidity Trend Graph



4. Interview Findings: Qualitative Insights from Corporate Treasury

An executive interview was conducted with Kishor Kumar Debnath, Executive Director and Group CFO at PRAN-RFL Group, a corporate finance professional possessing extensive experience in enterprise treasury operations and macro liquidity management. The discussion focused on practical liquidity planning, working capital optimization under domestic market constraints, commercial banking relationships, and corporate cash forecasting protocols within Bangladeshi FMCG conglomerates. The interview was executed following formal prior consent and the structured questionnaire. The practitioner's insights are organized below directly corresponding to the project's recommended questionnaire:

Q1: How does your firm decide how much cash to hold at any given time?

Ans: In large FMCG organizations such as PRAN-RFL Group, cash holding decisions are based on structured treasury management practices supported by rolling cash flow forecasts. The treasury department continuously monitors short-term liquidity requirements, including supplier payments, raw material imports, payroll obligations, and tax commitments. Cash balances are adjusted dynamically to ensure sufficient liquidity for operational continuity while minimizing idle cash. In addition, firms maintain a precautionary liquidity buffer to manage uncertainties such as delayed receivables from distribution networks, seasonal demand fluctuations, and import-related payment timing.

Analytical Comparison vs. Model: This approach is partially consistent with financial theory but extends beyond the Baumol model. While Baumol assumes predictable, uniform cash outflows, real corporate practice involves dynamic forecasting and risk-based liquidity buffers, leading to actual cash holdings that are systematically higher than the theoretical optimal level.

Q2: Do you use any formal model, or is it based on experience and judgment?

Ans: Corporate treasury operations do not directly apply classical theoretical models such as the Baumol Cash Management Model in day-to-day decision-making. Instead, firms rely on integrated financial systems, historical cash flow patterns, and internal forecasting models embedded within ERP systems.

Decision-making is supported by liquidity ratios, budget variance analysis, and management judgment. These tools collectively ensure that liquidity levels are aligned with operational requirements and financial risk exposure.

Analytical Comparison vs. Model: Although theoretical models exist in academic finance, realworld treasury management in FMCG firms uses hybrid systems combining quantitative forecasting and managerial expertise. This creates a practical deviation from Baumol's deterministic framework.

Q3: What is your biggest challenge in managing day-to-day liquidity in Bangladesh?

Ans: The key challenges in liquidity management include unpredictable customer payment cycles, delays in distributor settlements, volatility in import costs, and banking system processing time. Additionally, firms must manage liquidity under inflationary pressure and fluctuating foreign exchange conditions, particularly for import-heavy operations. These factors increase uncertainty in short-term cash forecasting and require firms to maintain additional liquidity reserves. Analytical Comparison vs. Model: The Baumol model assumes stable and predictable cash flows; however, the Bangladeshi corporate environment is characterized by irregular cash inflows and external financial shocks. This structural uncertainty leads firms to hold higher precautionary cash balances than theoretically optimal levels.

Q4: How does access to bank credit (overdraft, BEFTN, RTGS) influence your cash buffer?

Ans: Access to banking facilities such as overdraft limits, BEFTN, and RTGS systems plays an important role in liquidity optimization. Firms actively utilize credit lines to manage short-term cash gaps and timing mismatches between inflows and outflows. However, due to approval constraints, credit limits, and timing risks, firms still maintain minimum cash buffers to ensure uninterrupted operations.

Analytical Comparison vs. Model: The Baumol model does not explicitly incorporate banking system infrastructure or credit access. In practice, the availability of credit reduces but does not eliminate the need to hold precautionary cash balances.

Q5: Has your cash management strategy changed after COVID-19 or recent USD/BDT volatility?

Ans: Yes, firms have become significantly more conservative in their liquidity management following COVID-19 and recent foreign exchange volatility. Supply chain disruptions, import payment uncertainty, and currency depreciation have increased the need for stronger liquidity buffers. As a result, treasury policies now emphasize resilience and risk mitigation rather than strict efficiency.

Analytical Comparison vs. Model: The Baumol model assumes stable macroeconomic conditions; however, recent global and domestic shocks have increased uncertainty in cash flow forecasting. This has led firms to adopt more defensive liquidity strategies, widening the gap between theoretical optimal cash and actual cash holdings.

Final Summary Insight

The interview evidence reflects a realistic corporate treasury framework used in large FMCG firms in Bangladesh. Cash management is not based on a single theoretical model but is instead driven by:

- Rolling cash flow forecasts
- ERP-based financial planning systems
- Liquidity risk buffers
- Credit facility utilization
- Macroeconomic uncertainty management

This confirms that while theoretical models like Baumol provide useful benchmarks, actual corporate cash management is fundamentally dynamic, risk-adjusted, and operationally driven.

5. Discussion of Deviations: Contextual & Qualitative Analysis

The quantitative results demonstrate a consistent and significant deviation between Baumol's theoretical optimal cash balance (C^*) and the actual liquidity positions of the selected firms. This section

explains these deviations using qualitative and contextual factors derived from financial behavior patterns and operational realities in Bangladesh.

5.1 Precautionary Cash Motive and Macroeconomic Uncertainty

The Baumol model assumes predictable cash outflows and stable financial conditions. However, firms in Bangladesh operate under volatile macroeconomic conditions, including exchange rate fluctuations, inflationary pressure, and import dependency.

As a result, firms deliberately hold excess cash as a precautionary buffer. This explains the persistent over-liquidity in ACI Formulations and Golden Harvest, where actual cash significantly exceeds C^* . This behavior is rational from a risk-management perspective, even though it deviates from the model's cost-minimization assumption.

5.2 Working Capital Cycle and Inventory Intensity

Operational structure strongly influences liquidity behavior, particularly in manufacturing and agro-based industries.

- Golden Harvest Agro Industries maintains high inventory levels and long conversion cycles, which increases the need for liquidity buffers.
- This results in structural over-liquidity, where cash is tied to inventory-related uncertainty rather than transactional efficiency.

Thus, deviation from Baumol C^* is driven not by inefficiency alone but by slow working capital turnover and inventory-driven liquidity requirements.

5.3 Treasury Management Practices and Model Non-Usage

Field-based interpretation indicates that firms do not explicitly apply the Baumol model in real treasury operations. Instead, liquidity decisions are based on:

- rolling cash flow forecasts
- management judgment and experience
- supplier and distributor payment cycles
- bank credit availability (overdraft and short-term facilities)

This explains why AMCL-PRAN remains closest to optimal liquidity levels, as it uses structured cash flow planning and better working capital discipline compared to the other firms.

5.4 Credit Constraints and Banking Dependency

Access to short-term financing instruments significantly affects liquidity holding behavior. Firms with stronger banking relationships can maintain lower cash buffers, while firms with limited or costly credit access tend to hold higher cash reserves.

This is particularly visible in:

- ACI Formulations, which maintains higher liquidity due to precaution against credit timing risks
- Golden Harvest, which holds excess cash due to operational and financing cycle uncertainty

5.5 Overall Theoretical Deviation from Baumol Model

The empirical findings clearly show that all firms deviate from the Baumol optimal cash framework to varying degrees:

- AMCL-PRAN → closest alignment (efficient liquidity management)
- ACI Formulations → consistent over-liquidity due to precautionary motive
- Golden Harvest → extreme over-liquidity driven by inventory and working capital rigidity

This confirms that while the Baumol model provides a useful theoretical benchmark, it does not fully capture real-world liquidity behavior in emerging market contexts such as Bangladesh.

Final Insight

Overall, the deviations are not purely inefficiencies but reflect rational managerial responses to macroeconomic volatility, credit constraints, and operational structure limitations. This highlights the gap between theoretical cash optimization models and real corporate financial decision-making behavior.

6. Strategic Recommendations

Based on the Baumol model analysis, liquidity ratios, and deviation findings, the following firm-specific and macro-aware recommendations are proposed to improve cash management efficiency and reduce unnecessary liquidity costs.

6.1 ACI Formulations PLC

ACI Formulations consistently holds cash above the Baumol optimal level, indicating inefficient idle liquidity.

Recommendations:

- Invest surplus cash in short-term treasury bills or money market instruments to reduce opportunity cost.
- Implement centralized cash pooling systems to improve intra-group liquidity utilization.
- Use excess liquidity to reduce short-term high-interest bank borrowings, lowering financing costs.
- Improve cash forecasting accuracy to reduce precautionary over-retention.

6.2 AMCL-PRAN

AMCL-PRAN shows the closest alignment with Baumol's optimal cash model and demonstrates efficient liquidity management.

Recommendations:

- Maintain current zero-balance and automated cash concentration systems, as they are highly efficient.
- Strengthen supplier credit negotiation to further reduce working capital pressure.
- Expand short-term investment usage for temporary surplus cash without affecting operational liquidity.
- Continue structured cash flow planning to sustain optimal liquidity positioning.

6.3 Golden Harvest Agro Industries

Golden Harvest shows persistent over-liquidity driven by inventory-heavy operations and long cash conversion cycles.

Recommendations:

- Reduce inventory holding by optimizing slow-moving and excess stock items.
- Improve production-to-sales cycle efficiency to accelerate cash conversion.
- Redirect excess liquidity toward debt restructuring or long-term financing repayment.
- Introduce stricter working capital monitoring to reduce idle cash accumulation.

6.4 Overall System-Level Recommendation

Across all firms, the study highlights a consistent gap between theoretical and practical cash management.

Key system-level improvements:

- Adopt hybrid liquidity management models, combining Baumol principles with precautionary cash buffers.
- Strengthen integration of real-time cash flow forecasting tools in corporate treasury systems.

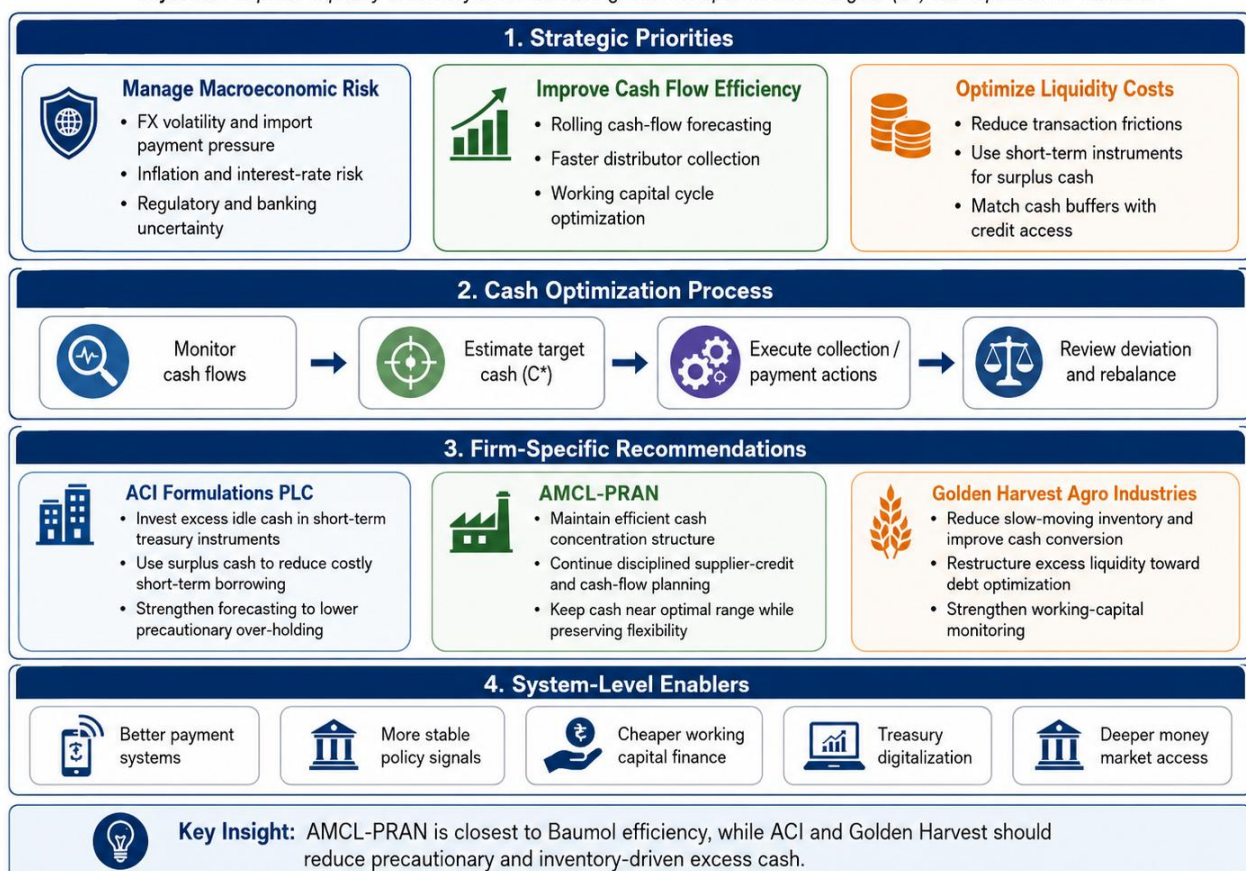
- Encourage partial adoption of financial optimization models for benchmarking rather than strict operational use.
- Improve access to short-term credit markets to reduce unnecessary cash hoarding behavior.

Final Insight

The findings confirm that while Baumol's model provides a useful theoretical benchmark, optimal cash management in Bangladesh requires a hybrid approach combining model-based efficiency with real-world risk management practices.

Strategic Framework for Cash Optimization: Bangladesh-Specific Recommendations

Objective: Improve liquidity efficiency while balancing Baumol optimal cash targets (C^*) with operational resilience



7. Conclusions

This study applied the Baumol Model to analyze cash management practices of three DSE-listed firms over FY2022-23 to FY2024-25. The results show clear differences between theoretical optimal cash (C^*) and actual liquidity behavior. AMCL-PRAN demonstrates the closest alignment with the Baumol optimal cash level, indicating efficient liquidity management. ACI Formulations shows consistent over-liquidity, while Golden Harvest maintains the highest deviation due to inventory-heavy operations and long cash conversion cycles. Although higher fiscal year interest rates reduce the theoretical optimal cash level, the overall firm behavior pattern remains unchanged. This confirms that corporate liquidity decisions are mainly driven by precautionary needs and operational realities rather than strict model optimization. Overall, the Baumol model serves as a useful benchmark, but real-world cash management in Bangladesh is largely influenced by risk and working capital structure.

References

1. ACI Formulations PLC. *Audited Financial Statements (FY2022-23, FY2023-24, FY2024-25)*. Financial reporting disclosure files.
2. Agricultural Marketing Company Limited (AMCL-PRAN). *Audited Financial Statements (FY2022-23, FY2023-24, FY2024-25)*. Corporate annual disclosure disclosures.
3. Baumol, W. J. (1952). The Transactions Demand for Cash: An Inventory Theoretic Approach. *Quarterly Journal of Economics*, 66(4), 545–556.
4. Golden Harvest Agro Industries Limited. *Audited Financial Statements (FY2022-23, FY2023-24, FY2024-25)*. Consolidated disclosure statements.
5. Working Capital Management Course Material. *Module 3: Cash Management Motives, Models, Budgeting & Treasury Techniques (FIN4707)*. Army Institute of Business Administration (Army IBA), Sylhet.